New forces on high-energy cosmic neutrinos: progress

Mauricio Bustamante

Niels Bohr Institute, University of Copenhagen

Progress report
October 31, 2026



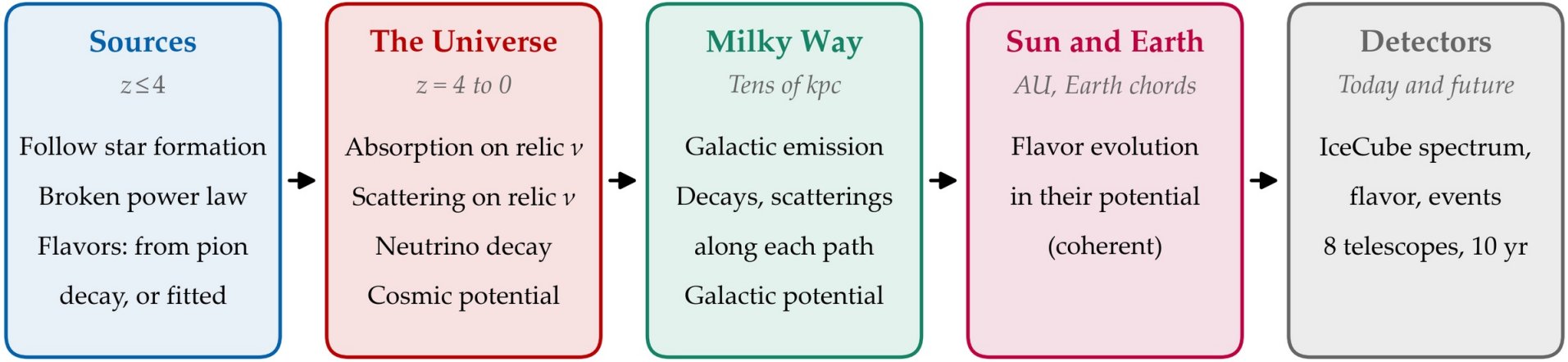


The idea: a new force acts on neutrinos on their way to us

IceCube sees cosmic neutrinos of 10 TeV to tens of PeV, from sources out to redshift ~4. On the way they cross the sea of relic neutrinos, the Milky Way, the Sun and the Earth.

A new, light mediator can **absorb** them, **scatter** them, make them **decay**, or **change how their flavors mix**, and so mark the energy spectrum or the flavor composition.

We compute all of these together, from the sources to the detector, and fit them to IceCube's data.



notes/notes.tex, Sec. "The transport equation in full"

Three families of models

Model	The mediator couples to	Matter potential from	Mediator masses
Dark portal (ν_e, ν_μ or ν_τ)	Neutrinos, through their mixing with a heavy neutrino: coupling $g_D \lvert U_{\alpha 4} \rvert^2$	None in these planes	10^{-30} eV – 1 GeV
$L_e - L_\mu$, $L_e - L_\tau$	Lepton-number differences: neutrinos and electrons, coupling g'	Electrons: cosmic, Galaxy, Sun, Earth	10^{-30} eV – 1 GeV
Secret interactions: scalar or vector; Majorana or Dirac neutrinos	Neutrinos only: ν_e, ν_μ, ν_τ, or all flavors equally	None	10^{-30} eV – 1 GeV

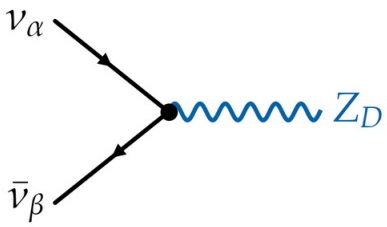
Each plane is one exclusion region, mediator mass against coupling. Every choice below is its own plane:

- **Dark portal and lepton number**: 5 models × 2 choices of source flavors (pion decay, or fitted) × 3 data sets = 30 planes below 10 keV, and 30 above.
- **Secret interactions**: 2 mediators (scalar, vector) × 2 kinds of neutrino (Majorana, Dirac) × 4 flavor choices × 2 mass ranges (below and above 1 keV) × 2 choices of source flavors × 3 data sets = 192 planes.

analysis/configs/campaign_ml, campaign_si · lepton-number models as in Agarwalla, Bustamante, Das, Narang, JHEP 08 (2023) 113

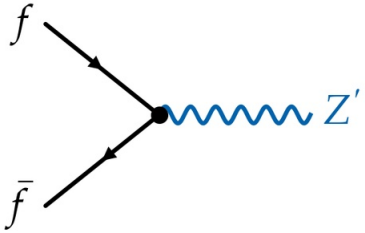
How the mediator couples

Dark portal



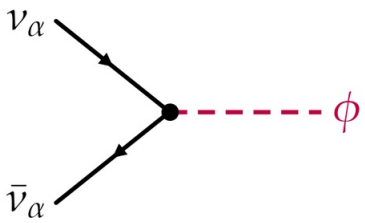
Coupling $g_D\, U_{\alpha 4} U^*_{\beta 4}$: through the mixing with a heavy neutrino, ν_4

$L_e - L_\mu$, $L_e - L_\tau$



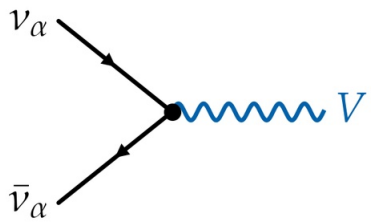
$+g'$ for ν_e, e; $-g'$ for ν_μ, μ (or ν_τ, τ)
Electrons: a matter potential

Secret interactions: scalar



Coupling g to ν_e, ν_μ or ν_τ, or to all equally; neutrinos only

Secret interactions: vector

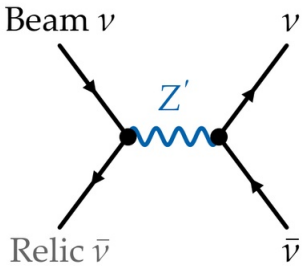


Coupling g to ν_e, ν_μ or ν_τ, or to all equally; its longitudinal mode couples as $m_\nu\, g/M$

The dark portal reaches the light neutrinos only through their mixing with a heavy neutrino, ν_4: its planes are drawn in the product $g_D\, |U_{\alpha 4}|^2$. The lepton-number boson also couples to electrons, which makes the matter potential. The scalar and the vector secret interactions are separate models, with planes of their own.

notes/notes.tex, Sec. “Matter charges and textures” (benchmark table)

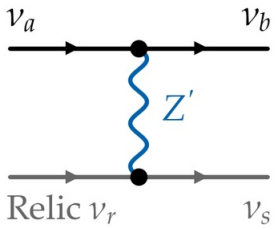
What the mediator does, by its mass



Resonant absorption

10 keV – 1 GeV

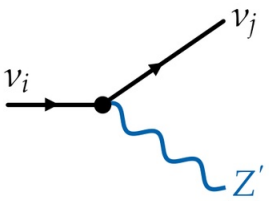
Peaks at $E = M^2/2m_\nu$: dips in the spectrum



Scattering on relics

Below 1 keV

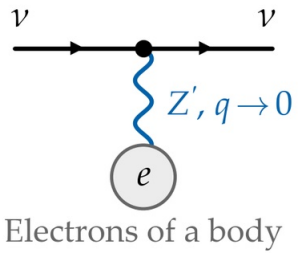
Changes the state: flavor change; energy lost if $r \neq s$



Decay

$M < m_i - m_j$

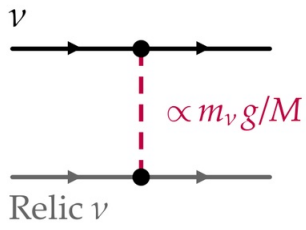
Heavier neutrinos decay: flux into the lightest state, lower energy



Matter potential

Light; $L_e - L_\mu$, $L_e - L_\tau$

Every electron within $1/M$ adds up: flavor mixing changes



Longitudinal mode

Below 1 keV; vectors

The vector acts like a scalar too: hard scatterings, absorption

All of these act together, in one equation (next slide). Grey lines: relic neutrinos, 56 per cm^3 per state today.

notes/notes.tex, Sec. “The transport equation in full”

The propagation equation, term by term

$F_a(E, z)$: neutrinos per unit energy in state a, an eigenstate of $H = UM^2U^\dagger/2E(1+z) + V$; E is the energy observed today. The same for antineutrinos. Solved from $z = 4$ to today: 439 energy bins, 200 redshift steps (400 where decays act), each bin's processes together and exactly.

$$\frac{\partial F_a(E,z)}{\partial t} = \underbrace{S_a}_{\text{Sources}} \underbrace{-\Gamma_a^{\text{abs}} F_a + \beta_a \int_E^\infty dE' \left[\frac{3E^2}{E'^3}\sum_c \Gamma_c^{\text{abs}} F_c + \frac{3(E'-E)^2}{E'^3}\sum_c \bar{\Gamma}_c^{\text{abs}} \bar{F}_c\right]}_{\text{Resonant absorption on the relic neutrinos, and what it re-emits}}$$

$$\underbrace{+\sum_{b\neq a}(R_{ab}F_b - R_{ba}F_a)}_{\text{Soft scatterings on the relics}} \qquad \underbrace{+\frac{\partial}{\partial E}(b_a F_a)}_{\text{Energy lost to the relics}}$$

$$\underbrace{-\Gamma_a^{\text{dec}} F_a + \sum_{a'}\int_E^\infty dE' \left[\frac{d\Gamma^{\text{keep}}_{a'\to a}}{dE} F_{a'}(E') + \frac{d\bar{\Gamma}^{\text{flip}}_{a'\to a}}{dE} \bar{F}_{a'}(E')\right]}_{\text{Decay into a lighter neutrino and the mediator, and the daughters}}$$

notes/notes.tex, Eq. (transport_cosmo) · code: src/dp_sweep.py, src/dp_medium.py

After the Universe: the Galaxy, the Sun and the Earth

Galactic neutrinos (CRINGE template) are emitted along each line of sight and carried to the Sun with the same processes:

$$-\frac{\partial G_a}{\partial \ell} = j_a + \text{ the same scattering, energy-loss and decay terms, at } z = 0$$

Extragalactic neutrinos crossing the Galaxy: there, the lightest neutrino and antineutrino states can decay into each other, which they cannot do between galaxies. So we follow them along every line of sight too.

Sun and Earth: no collisions or decays there, only coherent flavor evolution in the new potential, chord by chord through the Earth:

$$\frac{d\rho}{dx} = -i\,[H(x), \rho], \quad H(x) = \frac{UM^2U^\dagger}{2E} + V(x), \quad \rho = \sum_a (F_a + G_a)\,|a\rangle\langle a|$$

At the detector: the flux of each flavor, per band of declination or for the whole sky.

notes/notes.tex, Eq. (transport_gal) and Sec. “From the Sun to the detector” · standard matter effects left out (negligible at these energies)

In a potential, the states that travel change

The new potential is $V = AS$: A sets its size, S its flavor structure. For $L_e - L_\mu$, $S = \mathrm{diag}(1, -1, 0)$.

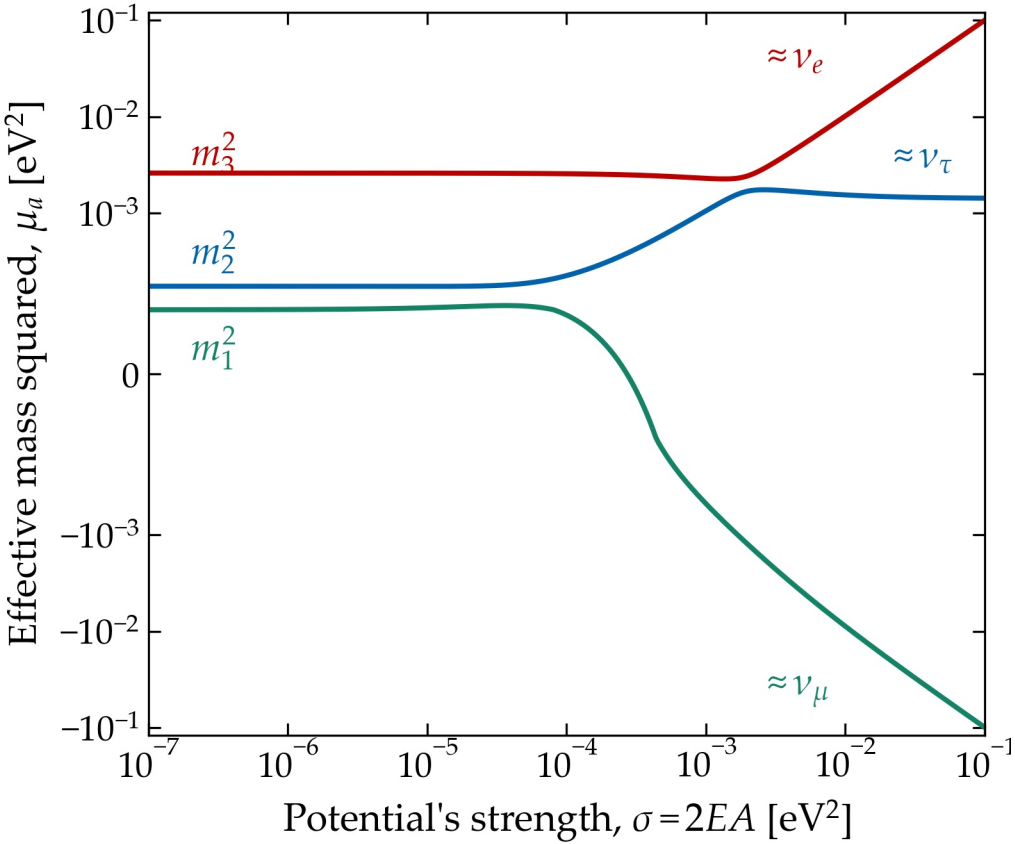
The states that travel are the eigenstates of $UM^2U^\dagger + \sigma S$, with $\sigma = 2E(1 + z)A$. Their effective masses squared are on the right.

Weak potential: they are the mass states, ν_1, ν_2, ν_3.

Strong potential: they become flavor states; ν_e and ν_μ get effective masses squared of about $+\sigma$ and $-\sigma$.

A decay $a \to b$ is open only where the effective masses allow it: the parent heavier than the daughter, with room for the mediator.

The old code decayed each state with vacuum masses, which let the most stable state decay into itself. Now each state decays with its own effective mass.



$L_e - L_\mu$, lightest mass 0.01 eV, NuFIT 6.1 · notes/notes.tex, "What is followed" and "Decay in a medium"

Three new effects

New effect	What it is	How much it matters
Decay inside a potential	In a potential, the states that travel are not mass states. Decays now go between these states, each with its own effective mass. The old treatment let a state decay into itself.	Galactic flux at 100 TeV, $L_e - L_\mu$ at $M = 10^{-16}$ eV: 3% of the standard one before, 57% now
Extragalactic neutrinos crossing the Galaxy	Inside the Galaxy, the lightest neutrino and antineutrino states can decay into each other. Between galaxies they cannot. We now follow these decays.	Lowers the flux at the top of the band by 13–41%, at $g/M = 0.1$ eV^{-1}
The vector's longitudinal mode	With massive neutrinos, the vector also scatters as a scalar would, with couplings $\propto m_\nu\, g/M$	Flux at 10 TeV × 0.95 (lightest mass 10^{-6} eV) or × 0.03 (0.1 eV), at $g/M = 0.1$ eV^{-1}

Also: each step is now solved exactly per energy bin, which cut the time per point from 600–750 s to 4–8 s.

notes/notes.tex, Sec. "Decay in a medium" · docs/dev/DECAY_IN_MEDIUM_2026-10-25.md

The data we fit

Data set	What it is	Role
IceCube spectrum + flavor	IceCube's 2026 combined fit of the astrophysical spectrum, in 1/3-decade segments from 4.6 TeV to 1 PeV; its 2025 flavor-composition measurement with medium-energy starting events (11.4 yr)	Main present-day result
IceCube events	HESE: 7.5 years of high-energy starting events. DR2: through-going muon tracks, IC79 + IC86, above 3 TeV. We fold our fluxes with IceCube's public detector responses.	Present-day, event by event, in energy and direction
Future telescopes	IceCube, IceCube-Gen2, KM3NeT/ARCA, Baikal-GVD, P-ONE, TRIDENT, NEON, HUNT: 10 years each; tracks, showers, double cascades	Projection, taking today's best fit as the truth

Atmospheric backgrounds: MCEq, Barr et al. uncertainties. The event set drops the flavor measurement, whose events overlap HESE's.

IceCube 2026: arXiv:2607.25966 · analysis/spectra.py, config.py · detectors as in NuFlavorSingleSource

Free parameters

Free parameter	Range or prior	Fitted in
Mediator mass M and **coupling**	On a grid; each mass read on its own	All fits
Astrophysical flux: normalization	0.01–1000 × IceCube's best fit; no prior	All fits
Spectral indices p, p_{low}, above and below the break at 25 TeV	2–3 and 0.5–2.5; no prior	All fits
Flavors at the sources, extragalactic and Galactic	Pion decay, (1/3, 2/3, 0); or (x, 1 − x, 0), x fitted	Half the planes
Galactic emission: normalization of the CRINGE template	Gaussian prior, 0.91 ± 0.18	All fits
Lightest neutrino mass (normal ordering)	10^{-6}–0.1 eV; sum of masses < 0.12 eV (95%)	All fits
Event fits only: atmospheric flux (12 Barr parameters, cosmic-ray index, prompt flux); HESE: two normalizations, three detector systematics; DR2: one normalization per configuration	Gaussian priors: the releases' for HESE, 10% for DR2	Events

Fixed: mixing angles and CP phase at the NuFIT 6.1 central values; sources follow the star-formation rate up to $z = 4$; as many neutrinos as antineutrinos.

analysis/configs/campaign_ml · analysis/events.py · NuFIT 6.1 · CRINGE: Schwefer, Mertsch, Wiebusch (2023)

How a limit is read

Frequentist: profile likelihood, $\chi^2 = -2 \ln L$.

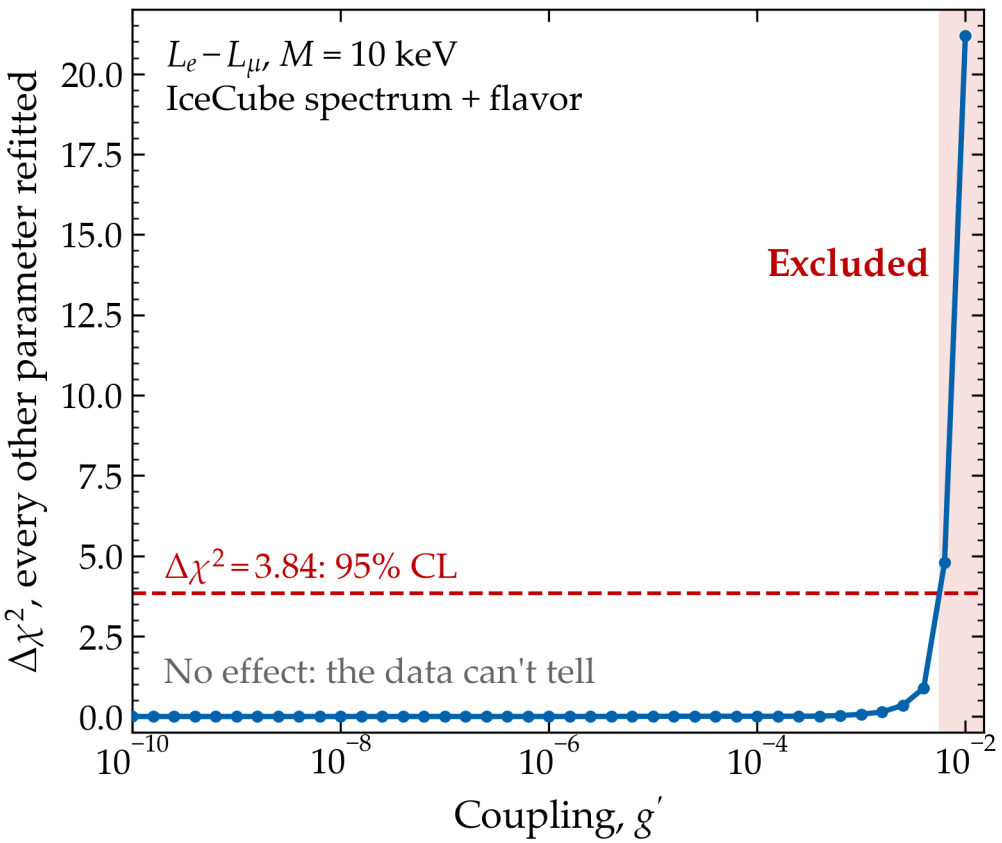
For each mediator mass, we scan the coupling; at each coupling, we refit every other parameter.

Excluded at 95% CL: $\Delta\chi^2 > 3.84$ above the best fit (or above the no-new-physics fit, where lower).

Small couplings change nothing: $\Delta\chi^2$ is flat. The limit is where it crosses 3.84.

In the likelihood: the data, plus the priors of the previous slide as Gaussian penalties. No other corrections. 3.84 is the two-sided value (one-sided: 2.71, tighter).

Planned: coverage tests with pseudo-experiments; Feldman–Cousins if the limits under-cover.



Right: $L_e - L_\mu$, $M = 10$ keV, IceCube spectrum + flavor, pion-decay flavors (the plane's own grid)

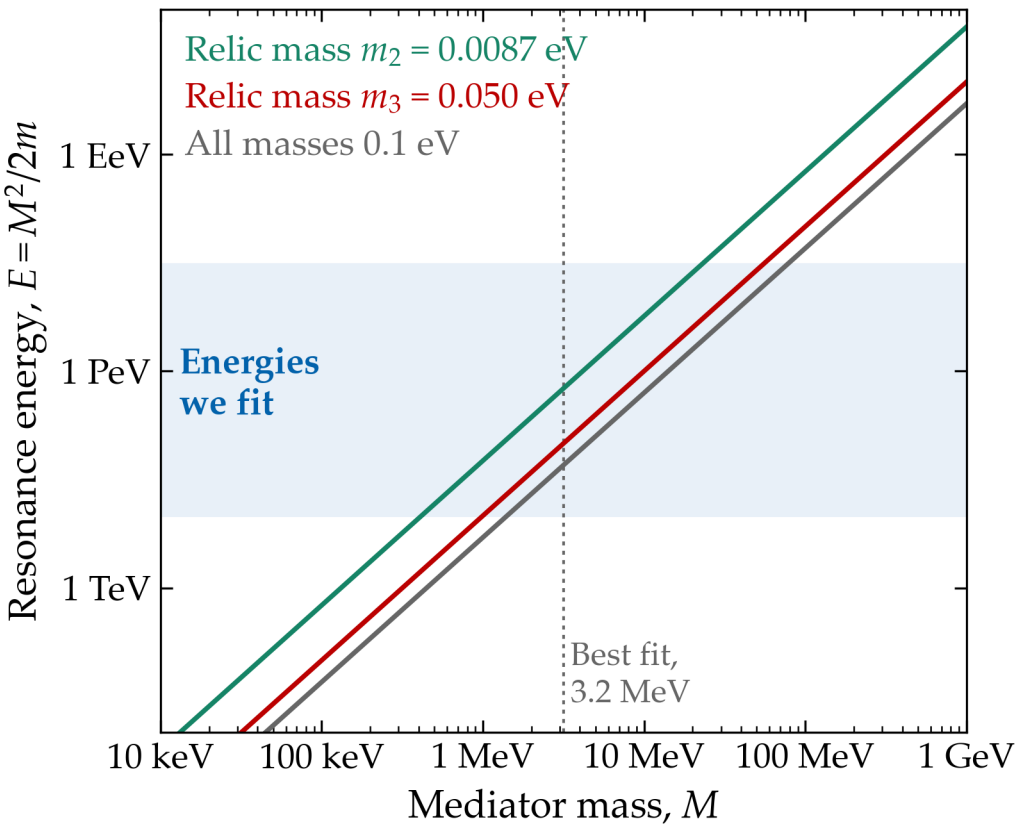
Why MeV mediators: where the resonance falls

A beam neutrino meets a relic neutrino of mass m, almost at rest. They make the mediator when $E = M^2/2m$.

For relics of $m_2 \approx 0.009$ eV to 0.1 eV and IceCube's energies, 10 TeV – 30 PeV, this needs $M \approx 0.4$–80 MeV.

There the absorption carves a dip in the spectrum, and the products pile up below it.

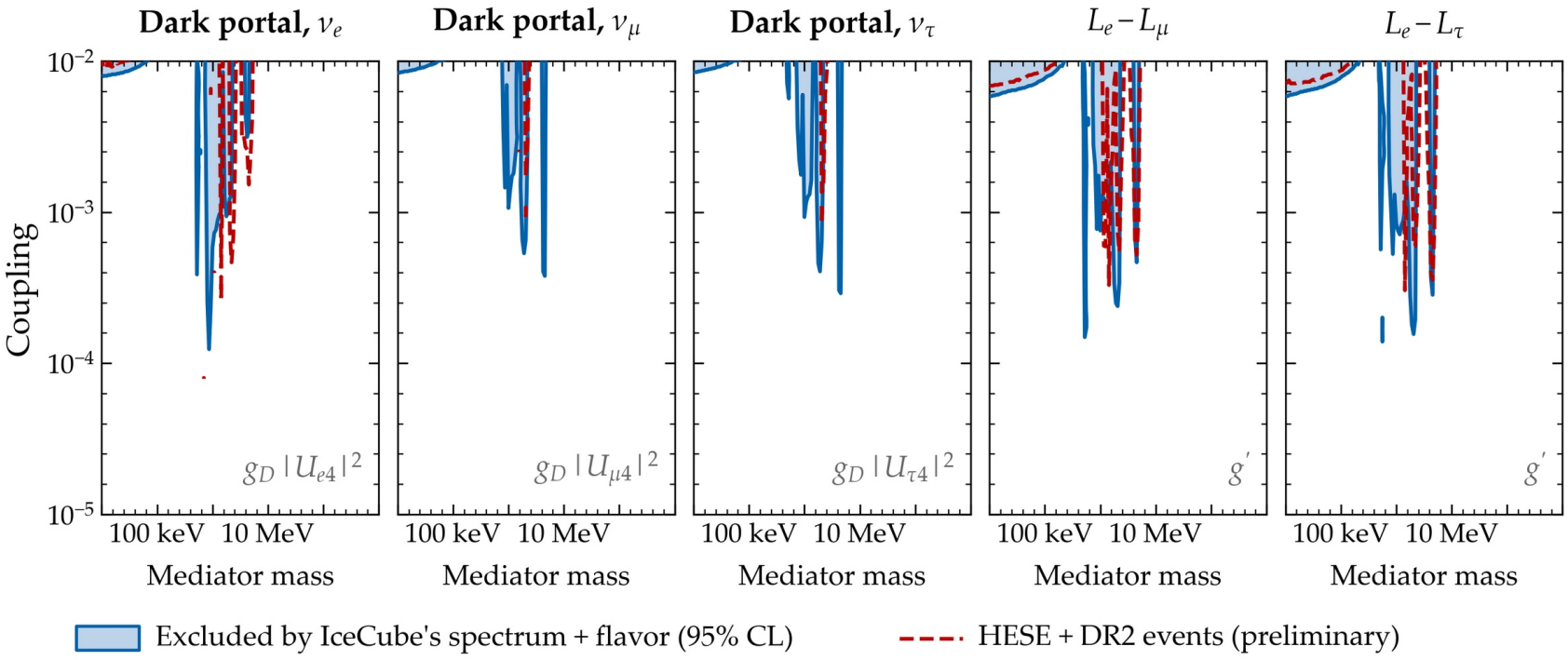
The data's best fit, $M \approx 3.2$ MeV, puts the resonance at ~100 TeV on m_3 and ~0.6 PeV on m_2.



Normal ordering, lightest mass ≈ 0 (NuFIT 6.1 splittings), or all masses 0.1 eV · notes, "Thermal absorption rates"

MeV mediators: what IceCube excludes today

Excluded: narrow windows near 0.5–5 MeV, where the resonance is in IceCube's band, down to couplings of 1.2–3.8 × 10^{-4}. IceCube's spectrum mildly prefers a resonance at $M \approx 3$ MeV ($\Delta\chi^2 = 5.7$–7.0 against no new physics).



Pion-decay flavors at the sources · unchanged by the October code

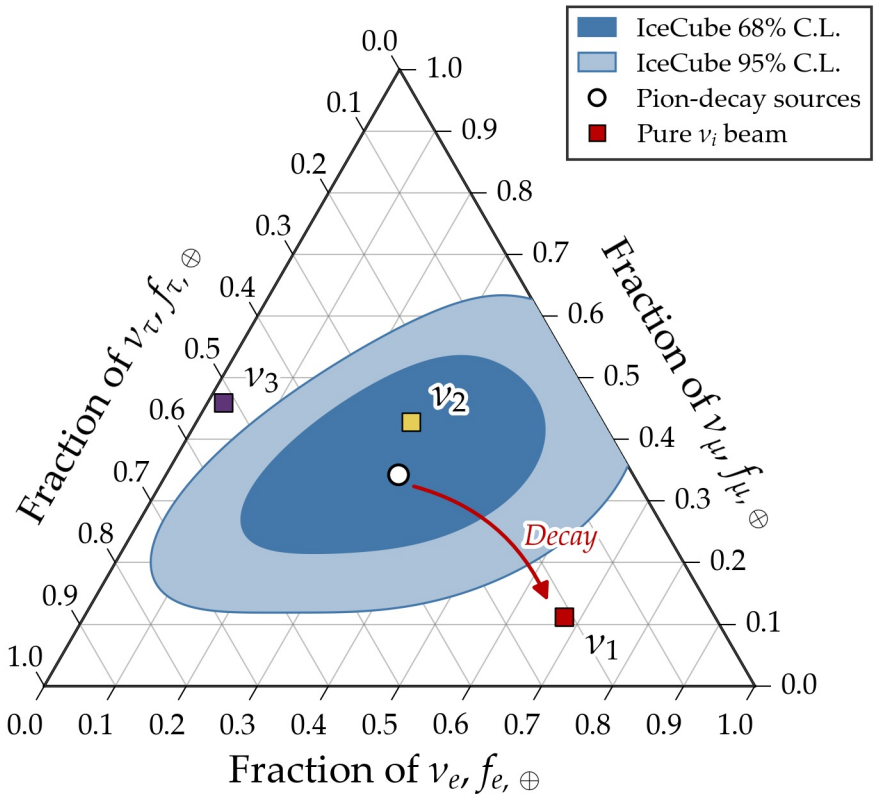
Ultralight mediators: first results (preliminary)

Dark portal, ν_μ, from IceCube's spectrum + flavor:

Mediator mass	Excluded $g_D \, \lvert U_{\mu 4} \rvert^2$
$\leq 10^{-10}$ eV	Outside the model's validity
10^{-8}, 10^{-6} eV	Every coupling $\geq 10^{-8}$
10^{-4} eV	$\geq 1.8 \times 10^{-8}$
10^{-2} eV	$\geq 1.8 \times 10^{-5}$
≥ 1 eV	$\geq 10^{-2}$

At light masses, every coupling in the plane makes the heavier neutrinos decay: the beam arrives as ν_1, mostly ν_e. IceCube's flavors disfavor that (right): $\Delta\chi^2 = 10$–11.

$L_e - L_\mu$, a first test at $M = 10^{-12}$ eV: $\Delta\chi^2 = 2.3$ at $g/M = 0.03$ eV^{-1}, 31 at 0.3 eV^{-1}.



Preliminary: one plane of the final run, limits not yet refined · flavors: IceCube's medium-energy starting events (2025)

Status and next steps

Step	Status
Dark portal + lepton number, MeV (30 planes)	**Done** · HESE + DR2 fits to be re-evaluated
Dark portal + lepton number, below 10 keV (30 planes)	**Running**: about half done at the last check (26 Oct); the cluster is offline now
Refine the limits, final plots	Next, when the cluster is back
Checks: mediator decaying back into neutrinos; relic population after decays; every vector point with $g/M \geq 10^{-2}$ eV^{-1} excluded	To do
Coverage tests with pseudo-experiments	Planned
Secret interactions (192 planes, 10^{-30} eV – 1 GeV)	Configured; runs after the ultralight run
One discovery case; then the paper	Last

Fits take 19–47 min per point with every process on (test of 2026-10-25)

Summary

One transport equation for everything a light mediator does to cosmic neutrinos: absorption, scattering, energy loss, decay, potentials. From the sources, through the Universe, the Galaxy, the Sun and the Earth, to the detector.

Three families of models, mediator masses 10^{-30} eV – 1 GeV, three data sets: today's IceCube spectrum + flavor, today's IceCube events, eight future telescopes.

MeV mediators: limits from IceCube's spectrum + flavor are done.

Ultralight mediators: the final run is more than half done. Where neutrinos decay, IceCube's flavors exclude even tiny couplings.

Next: finish the run, final plots, secret interactions, one discovery case, the paper.

Backup

Where we trust the models, and what we shade

Vector mediators below 1 keV: the vector's longitudinal mode couples as $m_\nu\, g/M$. At large g/M the theory breaks down (perturbative unitarity fails), and the heavier the neutrinos, the sooner. So at each point we fit only the lightest-neutrino masses where the theory holds. Where none does, we shade the point as not allowed.

Scalar secret interactions below 1 keV: the relic neutrinos themselves create a scalar field that shifts every neutrino mass. We do not model this. It matters above $g/M \approx 30$ to 10^4 eV^{-1}, depending on the flavor and the lightest mass; we shade that region.

Decays inside a potential: our decay rates assume each decay is slow against the oscillations between the states that travel. Every run records how well this holds. The correction is at most 1.5% at our test points; it reaches 10–50% only for $g/M \approx 10$–30 eV^{-1}, near where unitarity fails.

docs/SIMPLIFICATIONS_AND_ASSUMPTIONS.md, items 22–24 · docs/dev/REVIEW_2026-10-25.md · notes/notes.tex

Validation

Steps in redshift: where the potential sets the states, 200 steps converge to $\leq 1.3 \times 10^{-4}$ in spectral shape; where decays and scatterings cycle, 400 steps to $\leq 1.7 \times 10^{-3}$.

Coarse sky for the Galactic paths ($N_{\mathrm{side}} = 4$ vs 8): the crossing flux moves by 3×10^{-4}, the Galactic emission by 5.5% (its prior is 18%), the χ^2 by 0.1–0.3 at $\Delta\chi^2 \approx 30$.

Conservation: each decay keeps one neutrino. Without a potential, the vacuum decay rates come back exactly.

Nothing else moved: the MeV and secret-interaction planes above 1 keV are unchanged by the October code (952 test units, identical to 10^{-14}).

Review: an independent review of all October changes; the full test suite passes on the cluster and in continuous integration.

docs/dev/DECAY_IN_MEDIUM_2026-10-25.md · docs/dev/REVIEW_2026-10-25.md · notes/notes.tex

Backup: the terms in more detail

Sources: star-formation rate $R(z)$ up to $z = 4$, spectrum E^{-p}, flavors f_α split over the states at the source:

$$S_a = w_\nu \, \Phi_0 \, [E(1+z)]^{-p} \, R(z) \, (1+z) \sum_\alpha f_\alpha \, P_{\alpha a}(\sigma_{\mathrm{src}})$$

The states followed are the eigenstates of

$$H(E, z) = \frac{U \, \mathrm{diag}(m_1^2, m_2^2, m_3^2) \, U^\dagger}{2E(1+z)} + V(z)$$

Absorption: a Breit–Wigner on each relic mass state, averaged over the relics' thermal momenta; each absorption re-emits a neutrino and an antineutrino.

Soft scatterings: the conversion rate between states is damped where they decay fast (the decay widths enter the damping).

Decay in a medium, $a \to b$: allowed where $(1 - x)[x \, \mu_a(\sigma) - \mu'_b(x\sigma)] - x \, M^2 \geq 0$, with μ the effective masses squared and $x = E/E'$.

notes/notes.tex, Eqs. (sources), (transport_cosmo), (medium_window)